

Nitrate-N in Seawater

1 SCOPE

This method covers the determination of Nitrate-N in Seawater.

2 RANGE OF APPLICATION

Range 0.71 μM – 71.4 μM (0.01 – 1.0 mg N/L)

3 METHOD DETECTION LIMIT

By EPA Procedure MDL = 0.125 μM (0.002 mg N/L)

4 METHOD PRINCIPLE

The sample is mixed with pH buffer and then transferred to a copperized cadmium coil, where nitrate is chemically reduced to nitrite. The chemically-reduced sample is mixed with color reagent, prepared in dilute hydrochloric acid (or phosphoric acid). Original nitrite, plus nitrite from chemical reduction, reacts with sulfanilamide to form a diazonium compound. This species couples with N-(1-naphthyl)-ethylenediamine dihydrochloride to form a reddish-purple azo dye that is measured photometrically at 520 nm. Separate results for nitrite can be obtained using AQ method SEA-516. The AQ method EPA-516 is similarly equivalent to USEPA 353.2, except that cadmium reduction is omitted.

5 REFERENCES

Methods for the Determination of Inorganic Substances in Environmental Samples, EPA 600/R 93/100, August 1993: Method 353.2, Revision 2.0.

Standard Methods for the Examination of Water and Wastewater, APHA/AWWA/WEF, method 4500 NO_3^- F (2017 forward).

6 INTERFERENCES

Samples containing strong oxidants or reductants, such as chlorine or sodium thiosulfate, may cause low results by decreasing nitrate reduction efficiency. See section 11.1 for guidance on sample pre-treatment. Highly alkaline or overly-acidified samples may give low results due to the shift in pH. Neutralize the pH as needed.

Samples that contain oil and grease may coat the surface of the cadmium. This interference may be eliminated by sample pre-treatment with activated carbon or by extracting the sample with an organic solvent.

Bias from sample turbidity or color is correctable, using the sample blanking feature of the AQ software.

7 REAGENTS: CHEMICALS REQUIRED AND SAFETY INFORMATION

Consult the safety data sheets (SDS) for details on safe handling of chemicals.

<u>Chemical</u>	<u>Formula</u>	<u>C.A.S. No.</u>	<u>Safety Classification</u>
Copper(II) sulfate anhydrous or Pentahydrate	CuSO ₄ CuSO ₄ • 5H ₂ O	7758-95-7 7758-99-8	GHS07, GHS09 GHS07, GHS09
Hydrochloric acid, concentrated	HCl	7647-01-0	GHS06, GHS07
Imidazole	C ₃ H ₄ N ₂	288-32-4	GHS05, GHS07, GHS08
N-(1-naphthyl)-ethylenediamine dihydrochloride	C ₁₂ H ₁₄ N ₂ • 2HCl	1465-25-4	GHS07
Phosphoric acid, concentrated (low in nitrite)*	H ₃ PO ₄	7664-38-2	GHS05
Sodium hydroxide	NaOH	1310-73-2	GHS05
Sodium nitrate	NaNO ₃	7631-99-4	GHS03, GHS07
Sodium nitrite	NaNO ₂	7632-00-0	GHS03, GHS06, GHS09
Sulfanilamide	C ₆ H ₈ N ₂ O ₂ S	63-74-1	--
Triton® X-100	t-Oct-C ₆ H ₄ -(OCH ₂ CH ₂) _x OH x = 9-10	9002-93-1	GHS05, GHS07, GHS09

GHS01: Danger – Explosive

GHS04: Warning – Compressed Gas

GHS07: Warning – Irritant

GHS02: Danger – Flammable

GHS05: Warning/Danger – Corrosive

GHS08: Danger – Health Hazard

GHS03: Danger – Oxidizing

GHS06: Danger – Toxic

GHS09: Warning/Danger – Aquatic Toxicity

Note: Persons using this method should be familiar with normal laboratory practice. This method does not purport to address all of the safety problems, if any, associated with its use. It is the responsibility of the user to establish appropriate safety and health practices and to ensure compliance with local regulatory conditions.

*For alternative color reagent only

8 REAGENT SOLUTIONS

Use reagent-grade chemicals, certified for analytical or general laboratory use. SEAL recommends using ACS Reagent Grade whenever possible. Reagents of technical or commodity grade must be validated by the user. Use high-purity reagent water, distilled or deionized, and free from organic contamination, therefore Grade 1 (ISO Standard 3696) or better than Type II (ASTM Standard D1193) waters are suitable.

8.1 TRITON® X-100 7.5% (W/V)

Triton® X-100 (neat liquid)	15 g
DI Water	dilute to 200 mL

Tare a 400 or 600 mL beaker on a scale. Add 15 g of Triton® X-100 neat to the beaker. Add 150 mL of deionized water to the beaker and allow to stir. The Triton® X-100 will thicken initially, but will dissolve with continued stirring. Dilute to 200 mL total volume.

Stable for one month. Discard if solids are seen.

8.2 STOCK COPPER (II) SULFATE, 2% (w/v)

Copper (II) sulfate pentahydrate	31.3 g
or	
Copper (II) sulfate anhydrous	20 g
DI Water	dilute to 1 L

In a 1 liter volumetric flask, dissolve 31.3 g copper (II) sulfate pentahydrate or 20 g copper (II) sulfate, anhydrous in about 800 mL deionized water. Fill to the mark with deionized water and mix well.

Store at room temperature. This solution is commercially available. Stable for 6 months.

8.3 WORKING COPPER (II) SULFATE (0.01 M)

Stock copper (II) sulfate 2% (w/v) (above)	8.0 mL
DI Water	dilute to 100 mL

In a 100 mL volumetric flask, add 8.0 mL of stock copper (II) sulfate (Section 8.2). Dilute to the mark with deionized water.

Store at room temperature. This solution is commercially available. Stable for 6 months.

8.4 IMIDAZOLE BUFFER STOCK, PH 7.5 ± 0.05

Imidazole	16.3 g
Hydrochloric acid, concentrated, for pH adjustment	pH 7.5
DI Water	dilute to 1 L

Dissolve 16.3 g imidazole in about 800 mL deionized water. Let the solution get to room temperature, then adjust the pH to 7.5 ± 0.05 using hydrochloric acid, concentrated and dilute to 1 L.

Store at room temperature. Stable for 1 month.

8.5 WORKING IMIDAZOLE BUFFER

Imidazole buffer stock	150 mL
Working copper sulfate solution (0.01 M)	0.36 mL
DI Water	200 mL
7.5% (w/v) Triton® X-100 solution (above)	1.0 mL

To a 200 mL volumetric flask, add 150 mL imidazole buffer stock and 0.36 mL of working copper sulfate solution (0.01 M). Dilute to the mark with deionized water. Then add 1.0 mL 7.5% (w/v) Triton® X-100. Mix thoroughly.

Store at room temperature. Prepare fresh every week.

8.6 SULFANILAMIDE-NEDD

Hydrochloric acid, concentrated	15 mL
Sulfanilamide	2.0 g
Sodium hydroxide, pellets	1.9 g
N-(1-naphthyl)-ethylenediamine dihydrochloride	0.100 g
DI Water	dilute to 500 mL

Carefully add 15 mL concentrated hydrochloric acid to approximately 300 mL deionized water. Add 2.0 g sulfanilamide, 1.9 g of sodium hydroxide (pellets) and stir until dissolved. Add 0.100 g N-(1-naphthyl)-ethylenediamine dihydrochloride and stir until dissolved. Dilute to 500 mL with deionized water and store in a dark bottle.

Solution is stable for 1 month when refrigerated in a dark bottle. Discard when background absorbance of analyzed blanks exceeds 0.01 AU.

*** ALTERNATIVE RECIPE****

Sodium hydroxide	0.5 g
Phosphoric acid, concentrated	10 mL
Sulfanilamide	3.75 g
N-(1-naphthyl)-ethylenediamine dihydrochloride	0.19 g
DI Water	dilute to 500 mL

Dissolve 0.5 g sodium hydroxide pellets in about 250 mL deionized water. Carefully add 10 mL phosphoric acid, concentrated (low in nitrite). Add 3.75 g sulfanilamide and 0.19 g N-(1-naphthyl)-ethylenediamine dihydrochloride and stir to dissolve. Dilute to 500 mL with deionized water. Filter this reagent, e.g., using filter paper, such as Whatman #1 filter paper. As the reagent slowly turns pink, re-filter as needed.

Solution is stable for 1 month when refrigerated in a dark bottle. Discard when background absorbance of analyzed blanks exceeds 0.01 AU.

8.7 OPEN TUBULAR CADMIUM REACTOR COIL (OTCR)

The OTCR is a 14" coiled cadmium tube with a purple cladding. Within the OTCR, nitrate is chemically reduced to nitrite. There are different techniques for preparing your OTCR for use, which is dependent upon how often it is used.

NEW COIL

Perform an **Acid Strip** and **Install** the coil. In the MAINTENANCE menu, prime the syringe and **Regenerate** twice with 2% CuSO₄.

DAILY START-UP (when the coil is fluidically installed and recently used):

In the MAINTENANCE menu, prime the syringe and **Regenerate** once with water (instead of 2% CuSO₄).

IDLE COIL (when the coil has sat idle for 3 to 7 days):

Perform a **Water/Air Cleaning** and install the coil. In the MAINTENANCE menu, prime the syringe and regenerate twice with 2% CuSO₄.

LOW REDUCTION EFFICIENCY (or IDLE for 7+ days)

Follow NEW COIL instructions.

Installation

- The exposed ends of the coil are sleeved with Tygon® tubing, p/n 5719.
 - The tee connector on the rotating sampler arm connects to a 15 cm piece of 1/16" o.d. tubing.
 - Moisten the free end of this tube and insert it into the OTCR inlet sleeve, leaving a 1 mm gap from the cadmium coil end.
 - Locate the 1/16" o.d. dilutor transfer line, which passes beneath the sampler arm cosmetic cover. Insert the end of this tube into the other sleeve at the outlet of the OTCR.
- **Important**** Avoid making crimps or kinks to these 1/16" o.d. tubes.

Water/Air Cleaning

- Remove the coil by gently sliding the soft sleeve tubes off the cadmium ends.
- Use a 6" piece of Tygon® tubing to connect the male luer of a 50 mL syringe to the coil. SEAL recommends tubing p/n 116-0536-18, having 0.11" i.d.
- Using the syringe, rapidly draw deionized water into the coil, while purposely drawing air in alternation. Done vigorously, this technique expels the previous activated copper coating, seen as fine black flakes.

Acid Strip

- Remove the coil by gently sliding the soft sleeve tubes off the cadmium ends and attach the coil to the syringe as described above (Water/Air cleaning)
- Draw 6 N HCl into the coil and stand for 2–5 minutes.
- Perform **Water/Air Cleaning**

Regenerate using AQ Software

- In MAINTENANCE menu, select the Cadmium coil icon.
- Place copper sulfate (or water) in position 20; 50 or 100 ppm high standard nitrate in position 19; and working buffer in position 18.
- Click "continue" to run regeneration.

Determining Reduction Efficiency

To measure reduction efficiency, run a nitrite–N standard with the same concentration as a nitrate–N standard (normally mid-range). Reduction efficiency is calculated as follows:

$$\% \text{ Reduction Efficiency} = \frac{\text{N-NO}_3 \text{ Result}}{\text{N-NO}_2 \text{ Result}} \times 100\%$$

This reduction efficiency can be calculated in the software using the sample type REFF.

Refer to SEAL Analytical AQ Technical Tips series, Cd Coil Regeneration, for more information.

9 STANDARD SOLUTIONS

9.1 NITRATE-N STOCK STANDARD SOLUTION (35.7 mM / 500 mg N/L)

Sodium nitrate	3.034 g
Sulfuric acid, 5 normal (above)	up to 7 mL
DI water	dilute to 1 L

Dissolve 3.034 g sodium nitrate (dried at 105°C) in deionized water in a 1 liter volumetric flask. Up to 7 mL sulfuric acid, 5 normal, may be added for preservation. Dilute to the mark with deionized water.

Store up to 1 month at 4°C.

9.2 NITRITE-N STOCK STANDARD SOLUTION (3.57mM / 50 mg N/L)

Sodium nitrite	0.2462 g
DI water	dilute to 1 L

Dissolve 0.2462 g sodium nitrite (dried at 105°C) in deionized water in about 800 mL deionized water in a 1 liter volumetric flask. Dilute to the mark with deionized water.

Store up to 1 month in an amber bottle at 4°C.

9.3 CALIBRATION DILUENT (SYNTHETIC SEAWATER)

Sodium chloride	35 g
DI water	dilute to 1 L

Dissolve 35 g of sodium chloride in about 800 mL of deionized water in a 1 liter volumetric flask. Dilute to 1 liter with deionized water.

***Note:** The concentration of sodium chloride can be adjusted to match the salinity of the seawater samples. Low nutrient seawater may also be used and is commercially available.

Store at room temperature. Stable for 1 month.

9.4 AUTO CALIBRATION STANDARD – NITRATE-N (71.4 µM / 1.0 mg N/L)

NITRATE-N Stock Standard (35.7 mM, above)	0.5 mL
Calibration diluent	dilute to 250 mL

To a 250 mL volumetric flask pipet 0.5 mL NITRATE-N stock standard (35.7 mM / 500 mg N/L) and dilute to the mark with calibration diluent.

Prepare every two weeks.

9.5 MANUAL CALIBRATION

These are suggested concentrations for manual calibration standards.

Concentration, µM	0	0.357	1.785	3.57	7.14	17.85	35.7	53.55	71.4
Volume (mL) Auto Calibration Standard Solution diluted as indicated with calibration diluent	--	0.5	2.5	5.0	10	25	50	75	100
Final Volume, mL	100	100	100	100	100	100	100	100	100

Prepare weekly.

9.6 REDUCTION EFFICIENCY CHECK STANDARD – NITRITE-N (35.7 μ M / 0.5 mg N/L)

NITRITE-N Stock Standard (3.57 mM, above)	1.0 mL
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DI water	dilute to 100 mL
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To a 100 mL volumetric flask pipet 1.0 mL NITRITE-N stock standard (3.57 mM / 50 mg N/L) and dilute to the mark with deionized water.

Prepare twice weekly.

9.7 REDUCTION EFFICIENCY CHECK STANDARD – NITRATE-N (35.7 μ M / 0.5 mg N/L)

NITRATE-N Stock Standard (35.7 mM, above)	0.10 mL
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DI water	dilute to 100 mL
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To a 100 mL volumetric flask pipet 0.10 mL NITRATE-N stock standard (35.7 mM / 50 mg N/L) and dilute to the mark with deionized water.

Prepare twice weekly.

10 SAMPLE PRESERVATION AND STORAGE

Samples may be collected into plastic or glass containers.

The laboratory should consult the most currently acceptable sample storage and preservation procedures as mandated by the EPA.

11 PROCEDURE

Prepare standards and reagents as described in Sections 8 and 9. The laboratory must determine standard concentrations that are most suitable to their needs for calibration within the range of application for this method. The concentrations listed within this method are only suggestions.

Any reagents having visible particulates should be filtered.

Pour calibration standards, quality control solutions and samples into sample cups and reagent wedges, according to the user-defined work list.

Before running the analysis batch, check the analyzer for sufficient reagent water, fresh reaction segments, correct location of reagent wedges and a waste collection bottle in place.

11.1 CHEMISTRY TROUBLESHOOTING TIPS

Samples containing more than about 1 ppm residual chlorine require pre-treatment to prevent loss of nitrate reduction efficiency. Studies for nitrate reduction efficiency indicated that two drops of 1% (w/v) sodium sulfite (Na_2SO_3) effectively dechlorinated a 25 mL sample containing 5 ppm residual chlorine. Studies further indicated that adding 1% (w/v) sodium sulfite in excess of full dechlorination, between 1% and 6% volume-basis excess of the aliquot volume, caused a negative bias but did not cause persistent loss of coil reduction efficiency. In contrast, adding excess sodium thiosulfate, as suggested by USEPA Method 353.2, caused persistent loss of coil reduction efficiency.

12 TEST PARAMETERS

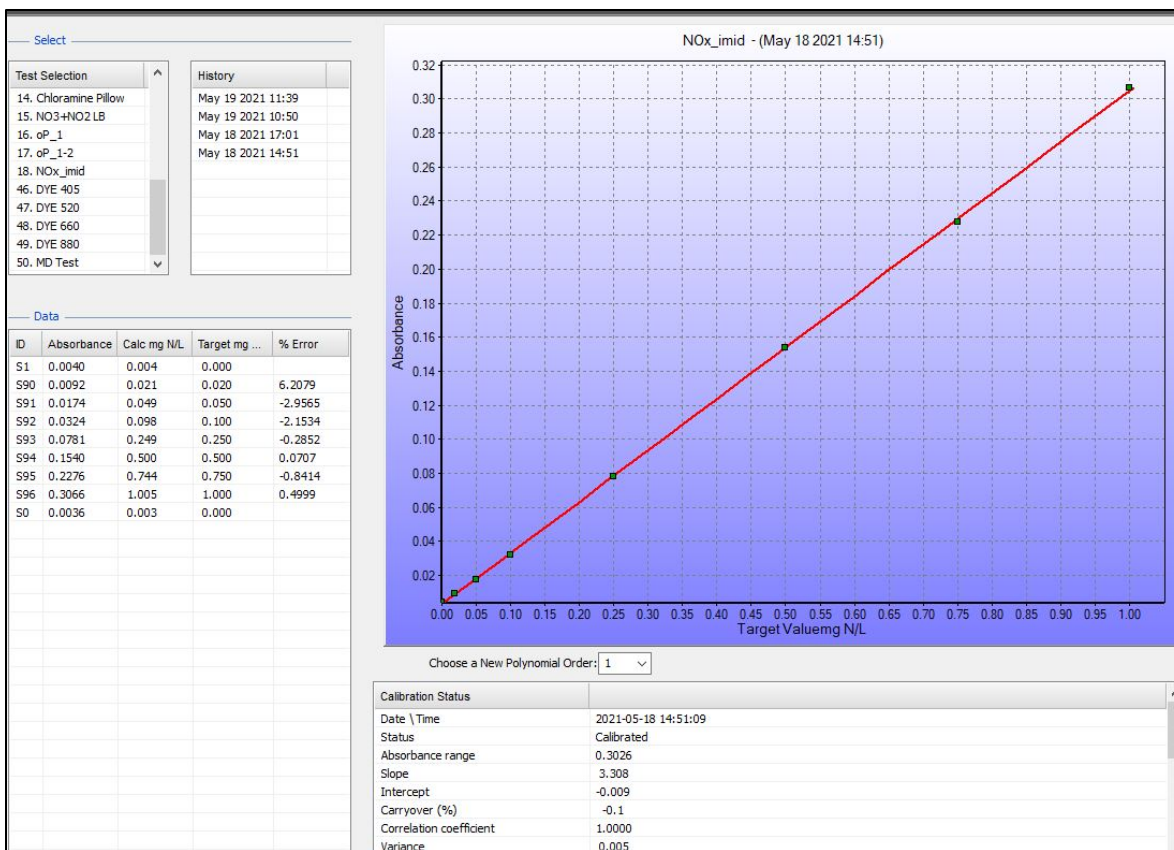
PARAMETER	AQ SETTING
Test name	NOx_1
Units	mg N/L
Decimals	3
Test type	Cadmium reduction
Sample volume (µL)	400
Water volume (µL)	100
Number of mixes	2
Cuvette primes	1
Cuvette washes	2
Reduction time (seconds)	25
Cadmium flush volume	280*
Reaction time (seconds)	480
Wavelength (nm)	520
Polynomial order	1
Number of reagents	2
1. Working Imidazole Buffer (µL)	160
2. Sulfanilamide – NEDD (µL)	350
Advanced Test Parameters	☒ Extra Debubbling Action

*Cadmium Flush Volume subject to slight changes, contact SEAL Tech Support for more information

DEFINITIONS:

Test name:	Name which appears on the final report
Units:	Concentration units appearing against the reported result
Decimals:	Number of decimal places in the reported result
Test type:	The type of test the software will perform
Sample volume:	Volume (in microliters) of the sample used in the test
Water volume:	Volume (in microliters) of water from the water reservoir used in the test
Number of mixes:	Number of mixes of the sample + reagent in the reaction well
Cuvette primes:	Number of times the cuvette is flushed with reagent (no sample) prior to sampling
Cuvette washes:	Number of washes of the cuvette between aspirations of finished reaction liquid
Baseline on Wash:	Water baseline measured between aspirations to provide drift correction
Reaction time:	Time (in seconds) between addition of sample + reagent and measurement of the reaction mixture
Wavelength:	Interference filter wavelength to be used in the test
Polynomial order:	Polynomial order to be used when fitting the standard points to the standard graph (1 = linear)
Number of reagents:	Number of reagents defined for the test
Reagent volume:	Volume (in microliters) of the reagent used in the test

13 STANDARDIZATION



Max. Blank Absorbance (S1): 0.01 AU

Sensitivity Range (71.4 μM / 1.00 mg N/L): 0.25 – 0.37 AU

DEFINITIONS:

Abs. Range: Difference between the absorbance of the highest standard and the absorbance of the blank

Variance: Calculated as follows:

$$\text{Variance} = \frac{\sum(\text{Deviations, i.e., errors})^2}{\sum(\text{Observed values})^2} \times 100$$

Carryover (%): Calculated as follows:

$$\text{Carryover (\%)} = \frac{(A - B)}{C} \times 100\%$$

Where,

- A = absorbance of carryover blank (S0)
- B = absorbance of blank (S1)
- C = calibration absorbance range

14 PERFORMANCE VALIDATION DATA

14.1 DETECTION LIMIT STUDY

Replicates of 0.71 μM (0.01 mg N/L) nitrate standard in synthetic seawater***

	Concentration (μM)			
	0.79	0.79	0.79	0.71
	0.71	0.71	0.86	0.64
	0.86	0.71	0.86	0.71
	0.79	0.79	0.93	0.71
	0.79	0.79	0.86	0.71
	0.64	0.79	0.86	0.71
	0.50	0.71	0.93	0.71
	0.71	0.86	0.86	0.71
	0.57	0.79	0.86	0.71
	0.57	0.79	0.86	0.71
	0.57	0.79	0.86	0.64
	0.71	0.79	0.93	0.79
	0.79	0.79	0.86	0.71
	0.93	0.79	0.86	0.79
	0.79	0.79	0.86	0.71
Statistics				
Mean	0.71	0.78	0.87	0.71
Std. Deviation	0.12	0.04	0.04	0.04
MDL (μM)*	0.32	0.10	0.10	0.10
MDL ($\mu\text{g N/L}$)	0.004	0.001	0.001	0.001

14.2 PERFORMANCE STUDY

Replicates of 3.57 μM (0.05 mg N/L) nitrate standard in synthetic seawater***

	Concentration (μM)			
	3.57	3.64	3.36	3.64
	3.64	3.64	3.36	3.79
	3.79	3.50	3.43	3.71
	3.93	3.79	3.43	3.71
	3.86	3.64	3.21	3.79
	3.71	3.57	3.43	3.64
	3.71	3.64	3.43	3.79
	3.57	3.57	3.43	3.86
	3.71	3.57	3.43	3.93
	3.57	3.64	3.36	3.79
Statistics				
Mean	3.71	3.62	3.39	3.76
Std. Deviation	0.12	0.08	0.07	0.09
% RSD**	3.3%	2.1%	2.0%	2.4%
% Recovery	103.8%	101.4%	94.8%	105.4%

14.3 PRECISION STUDY

Replicates of 3.57 μM (0.50 mg N/L) nitrate standard in synthetic seawater***

Concentration (μM)				
Statistics				
Mean	35.59	34.84	34.64	35.39
% Recovery	99.7	97.5	97	99.1
Std. Deviation	0.69	0.53	0.77	0.94
% RSD**	2.0	1.5	2.2	2.7
Maximum	36.29	35.93	35.50	37.14
Minimum	34.64	34.14	33.14	33.93

Replicates of 71.42 μM (1.00 mg N/L) nitrate standard in synthetic seawater**

Concentration (μM)				
Statistics				
Mean	69.21	71.06	70.04	71.41
% Recovery	96.9	99.5	98.1	100.0
Std. Deviation	1.98	2.09	1.28	2.13
% RSD**	2.9	2.9	1.8	3.0
Maximum	71.50	73.07	72.29	75.57
Minimum	64.64	65.79	68.21	68.14

* MDL is calculated by multiplying the Standard Deviation by the Student-t value for the number of replicates run.

** %RSD is calculated by dividing the Standard Deviation by the mean, then multiplying by 100.

*** To prepare these standards, refer to Section 9. Synthetic seawater contains 35 g/L of NaCl.

15 REVISIONS

June 2021

Rev 0

Method created and support data acquired